

Protocols — Guided Notes ANSWER KEY

Unit 1.5.2 | CompTIA Network+ N10-009 Obj. 1.5 | N10-008 1.5

For instructor use / distribute after students complete the notes.

Question 1

ICMP does not carry _____ data. It operates at Layer _____ and does not use _____. The two most common uses of ICMP in troubleshooting are _____ and _____.

Answer: User; Layer 3; port numbers; ping (Echo Request/Reply); traceroute/tracert (Time Exceeded messages).

Question 2

If ping fails to a host that is actually up and running, the most likely reason is _____. This means a failed ping does NOT automatically mean _____.

Answer: ICMP is being filtered/blocked by a firewall or security policy; the host is down.

Question 3

ARP resolves _____ addresses to _____ addresses. A device broadcasts an ARP _____ to the segment, and the target responds with a unicast ARP _____. The result is stored in the device's _____ table.

Answer: IP; MAC; Request; Reply; ARP cache.

Question 4

Write out the DHCP DORA process:

D — _____

O — _____

R — _____

A — _____

Answer: D — Discover (client broadcasts looking for a DHCP server); O — Offer (server offers an IP address and config); R — Request (client formally requests the offered address); A — Acknowledge (server confirms the lease).

Question 5

A rogue DHCP server occurs when _____. The symptom on affected workstations is _____. To find the rogue server, you would _____.

Answer: An unauthorized device (e.g., a home router plugged into the network) starts responding to DHCP Discover broadcasts; devices receive incorrect IP configurations (wrong gateway, wrong DNS) and lose connectivity; check DHCP server logs, use a packet capture to see which MAC address is sending DHCP Offers, or enable DHCP snooping on managed switches.

Question 6

In the DNS resolution process, if the local DNS server doesn't have a cached answer, it queries a _____ name server first, which directs it to the _____ server for the domain's extension, which directs it to the _____ name server that has the actual IP address.

Answer: Root; TLD (top-level domain, e.g., .com); authoritative.

Question 7

Before HTTP data moves over HTTPS, the client and server perform a _____ handshake. The server presents a _____ to prove its identity. All traffic after this point is _____ on port _____.

Answer: TLS; certificate; encrypted; 443.

Question 8 — Real World Application

REAL WORLD: A user reports 'the internet is down.' You run ping 8.8.8.8 — it succeeds. You run ping google.com — it fails. Walk through your diagnosis and fix.

Answer: IP connectivity is working (ping by IP succeeds), but DNS resolution is failing (ping by name fails). Diagnosis: check the workstation's configured DNS server (ipconfig /all). Try nslookup google.com — if it times out or returns a server failure, the DNS server is unreachable or not functioning. Fix options: confirm the DNS server IP is correct (not a DHCP misconfiguration), check if the DNS server is running, check firewall rules for UDP/TCP port 53 between the client and DNS server, or temporarily configure a known-good DNS server (8.8.8.8) to confirm the fix.